

- 8 (a) Apart from being different types of forces, state a difference between electric force and magnetic force.

Electric forces are (created by and) act on, both moving and stationary charges; while magnetic forces are (created by and) act on only moving charges.

[1]

- (b) Two oppositely charged parallel metal plates P and Q are placed in a vacuum. The electric field is uniform in the region between the plates.

A uniform magnetic field also exists in the region between the plates. The direction of the magnetic field is into the page as illustrated in Fig. 8.1.

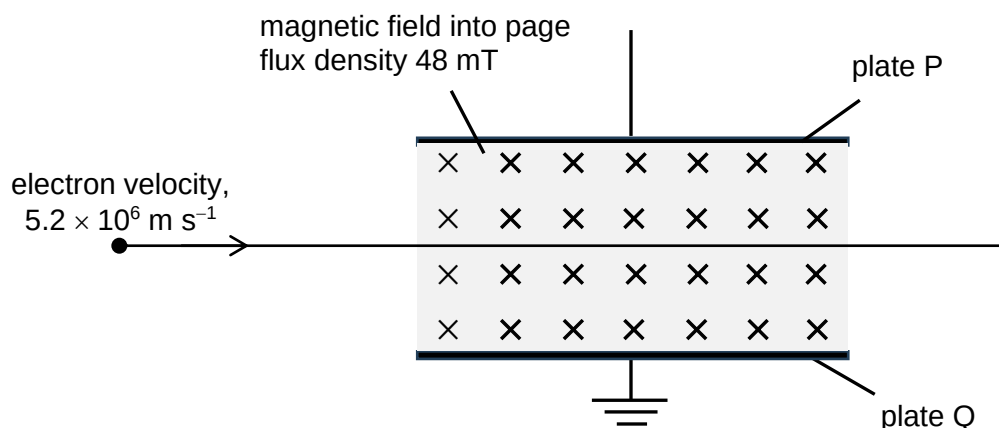


Fig. 8.1

An electron enters the region between the plates at right angles to both the electric field and the magnetic field. The electron travels through the field.

The magnetic flux density is 48 mT. The velocity of the electron is  $5.2 \times 10^6 \text{ m s}^{-1}$ .

The magnetic force and electric force acts on the electron in opposite directions.

- (i) State and explain the polarity, positive or negative, of plate P.

By Flemings left hand rule, the magnetic force on the electron acts downwards. [M1]  
Hence the electric force on the electron should act upwards, and plate P will be positive with respect to plate Q. [A1]

[2]

- (ii) Calculate the magnitude of the magnetic force  $F_M$  acting on the electron.

$$\begin{aligned} F_M &= Bqv \\ &= (48 \times 10^{-3})(1.6 \times 10^{-19})(5.2 \times 10^6) \quad \text{M1} \\ &= 4.0 \times 10^{-13} \text{ N} \quad \text{A1} \end{aligned}$$

$$F_M = \dots\dots\dots \text{ N [2]}$$

- (iii) The electron passes through the field undeflected when the magnitudes of the magnetic and electric forces are the same.

With reference to the forces acting on the electron as it passes through the plates, state and explain how the path will change if the potential across the metal plates is decreased slightly.

When the potential decreases, the electric field strength and hence the electric force on the electron decreases. ( $F = qE$ ) [B1]

Upon entering the field, the downwards magnetic force is larger than the upward electric force and hence there is a net downward force. The electron is deflected downwards. [A1]

As the electron path curves downwards, the subsequent net force is the resultant of the upward electric force and the magnetic force tangential to the motion of the electron. The path curves further downwards. [B1]

[3]

- (c) Two flat coils, P and Q each of diameter 29 cm are fixed so that their planes are parallel and are separated by a constant distance equal to the radius of each coil, with the direction of the current as shown in Fig. 8.2.

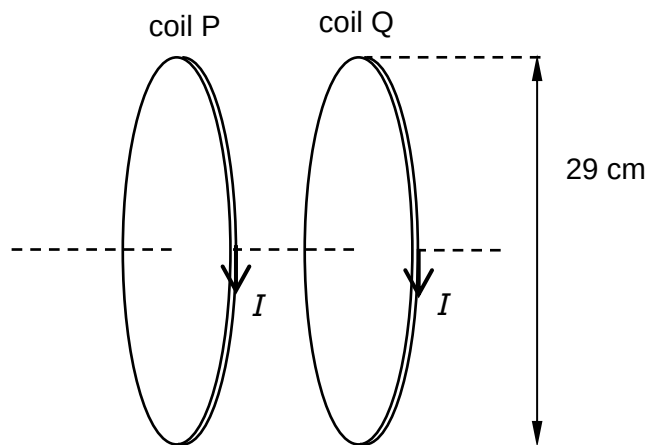


Fig. 8.2

The current  $I$  in both coils is 1.3 A.

The magnetic flux density  $B$  in the region between the two coils is uniform and given by the expression

$$B = 0.72 \mu_0 \frac{NI}{r}$$

where  $N$  is the number of turns on each of the flat coil of radius  $r$ . The permeability of free space is  $\mu_0$ .

- (i) Explain how a uniform field is set up between the coils.

When there is a current in the coil, by right hand grip rule, there is field set up around each loop such that the magnetic flux passes from left to right. [B1]

The fields from each of the loops superpose to set up a close to uniform field between them.

[2]

- (ii) Each coil has 160 turns. Show that the magnetic flux density  $B$  is approximately 1.3 mT.

$$\begin{aligned}
 B &= 0.72 \mu_0 \frac{NI}{r} = (0.72)(4\pi \times 10^{-7}) \frac{(130)(1.3)}{(2.9 \times 10^{-2}/2)} \\
 &= 1.298 \times 10^{-3} \\
 &= 1.3 \text{ mT}
 \end{aligned}$$

[1]

- (iii) The space between the coil in (c) is a vacuum.

An electron of velocity  $5.2 \times 10^6 \text{ m s}^{-1}$  travels at right angles into the uniform field produced by the two coils.

Calculate the radius of its orbit in the magnetic field.

Magnetic force provides centripetal force [B1]

$$\begin{aligned}
 \sum F &= ma_c \\
 F_M &= Bqv = \frac{mv^2}{r} \\
 (1.3 \times 10^{-3})(1.6 \times 10^{-19})(5.2 \times 10^6) &= \frac{(9.11 \times 10^{-31})(5.2 \times 10^6)^2}{r} \quad \text{M1} \\
 r &= 2.2775 \times 10^{-2} \\
 &= 2.28 \text{ cm} \quad \text{A1}
 \end{aligned}$$

radius = ..... m [3]

- (d) The magnetic field in (c) is rotated. The initial direction of the electron is now at an angle of  $30^\circ$  to the direction of the uniform magnetic field, as shown in Fig. 8.3.

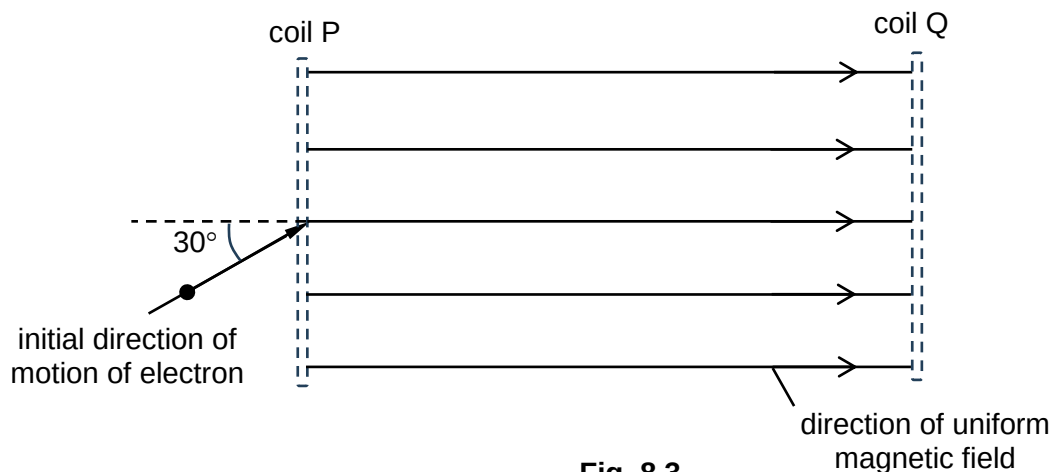


Fig. 8.3

- (i) State the path of the electron in the magnetic field.

The electron moves in a helical path.

[1]

- (ii) By considering the components of the velocity parallel to the magnetic field and at right angles to the magnetic field, explain the motion of the electron as stated in your answer in (d)(i).

The horizontal component of velocity is parallel to the direction of the B field, hence remains unchanged. [B1]

By Flemings left hand rule, a magnetic force acts on the electron perpendicular to the B field and its perpendicular component of velocity. This provides the centripetal force causes the electron to chart out a circular path in the plane perpendicular to the B field.

The motion of the electron is the vector sum of the horizontal and perpendicular components of velocity, hence a helical path. [A1]

[3]

- (iii) State and explain how the path of the electron will change if the current  $I$  in the coils were increased.

When current  $I$  is increased, the magnetic flux density  $B$  increases.

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

Since  $m$ ,  $v$  and  $q$  are constants. [M1]

[2]

This causes the radius  $r$  of the helical path to decrease [A1]

[Total: 20]

**End of Paper**